
PG-MoE: Physics-Driven Phase-Gated Mixture-of-Experts for RGB-IMU Activity Recognition

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Abstract

Multimodal fusion for human activity recognition (HAR) typically applies fixed or learned weights to combine modalities, without accounting for how the informational dominance of each modality shifts across biomechanically distinct action phases. We propose PG-MoE (Phase-Gated Mixture-of-Experts), a framework that uses physics-derived features from inertial measurement unit (IMU) signals—acceleration magnitude, jerk, and kinetic energy rate—to produce interpretable, per-timestep gating weights that dynamically route between an RGB vision expert and an IMU expert. We evaluate on UTD-MHAD using RGB and IMU modalities, a more practical but challenging setting than prior skeleton-based approaches. Our vision expert (ResNet3D, fine-tuned) achieves 89.3% and our IMU expert (deep 1D-CNN) achieves 82.3% standalone. We compare nine fusion strategies: Late Fusion achieves the best accuracy at 92.3%, while PG-MoE with expert fine-tuning reaches 91.4% with the added benefit of interpretable temporal gating. An oracle analysis reveals a 98.4% upper bound, confirming strong but underutilized modality complementarity. We additionally show that frozen vision-language model features (Qwen2.5-VL, 58.6%) substantially underperform task-specific video models for HAR, highlighting the importance of motion-aware pretraining.

1 Introduction

Recognizing what a person is doing from sensor data underpins many systems we interact with daily, from fitness trackers and rehabilitation monitors to gesture-controlled interfaces. The accuracy of these systems depends on the sensors that drive them. RGB cameras provide rich spatial and contextual information, but degrade under motion blur, occlusion, or rapid out-of-frame motion. Body-mounted inertial measurement units (IMUs) capture high-frequency motion signatures directly at the source, but lack scene context and often confuse actions with similar kinematic profiles. Fusing video and IMU is a natural response to these complementary weaknesses, and it has been the subject of a substantial body of prior work.

Across existing fusion approaches, the gating signal between modalities, when one is used at all, is typically learned from latent features through cross-modal attention, or held fixed across the action by averaging logits or concatenating streams. These designs do not explicitly tie their fusion weights to the physical structure of the action being recognized, even though many activities have a clear biomechanical structure. In a tennis serve or a baseball swing, the preparation phase carries pose and scene cues that video reads well, while the high-velocity execution phase produces sharp accelerometer transients that the IMU captures more directly. A gating mechanism that consumes only latent features may eventually learn this behavior, but it does not make the structure visible to inspection.

In this paper, we propose PG-MoE, a phase-gated mixture-of-experts framework for HAR with RGB and IMU. The central design choice is to predict the per-token modality-fusion weight $\alpha(t)$ directly from biomechanical quantities derived from the raw IMU signal (acceleration magnitude, a jerk-like second derivative, and a kinetic-energy rate), rather than from learned latent features. The resulting Phase Arbitrator has 2,881 trainable parameters and produces a gating signal in interpretable physical units. We make three contributions. First, we introduce this physics-grounded per-token arbitrator and combine it with a deep 1D-CNN IMU expert (82.3% standalone on UTD-MHAD) and a fine-tuned ResNet3D vision expert (89.3% standalone). Second, we compare nine fusion strategies across two vision backbones, a frozen vision-language model (Qwen2.5-VL, 58.6%) and a fine-tuned task-specific 3D-CNN (ResNet3D, 89.3%), and the gap between them reveals how much the choice of vision backbone matters in this setting. Third, an oracle analysis shows that perfect per-sample modality routing would reach 98.4% accuracy, well above the 92.3% of a plain late-fusion baseline and the 91.4% of PG-MoE. We use this gap to discuss when phase-gated fusion is most useful and when, on small datasets, simple averaging is hard to beat.

2 Related Work

2.1 Multimodal fusion for activity recognition

Foundational surveys of multimodal machine learning identify reasoning across modalities, defined as deciding which modality to trust at each moment, as one of the open challenges in the field Liang et al. [2022]. Benchmarking efforts across many datasets and modalities consistently report that modality robustness, including how a model behaves when one stream is degraded or missing, remains a primary source of difficulty for current systems Liang et al. [2021]. Dynamic Multimodal Fusion (DynMM) addresses this with a learnable gating network that selects an inference path per sample Xue and Marculescu [2023], while MMTSA introduces a segment-level attention mechanism that weights different temporal segments of a concatenated RGB+IMU representation Gao et al. [2023]. Closer to our setting, DG-IVF proposes a dual-gated architecture specifically for IMU and vision fusion in HAR Li et al. [2023], and Centaur emphasizes robustness to modality dropout in wearable settings Xavier et al. [2024]. A complementary line of work uses knowledge distillation: XTinyHAR distills a multimodal teacher into an IMU-only student to retain accuracy with reduced inference cost Lamaakal et al. [2025]. Across these methods, the fusion weight is either learned from latent features or kept constant within a sample, and the gating decision operates at sample or segment granularity.

2.2 Vision backbones for action recognition

Video backbones commonly used in HAR fall into three broad families. The first is task-specific 3D convolutional architectures pretrained on large action datasets such as Kinetics-400 Kay et al. [2017]. I3D introduced the inflation paradigm that allows 2D ImageNet weights to be reused for 3D action recognition Carreira and Zisserman [2017], and subsequent designs including R(2+1)D / r3d_18 Tran et al. [2018a] and SlowFast Feichtenhofer et al. [2019] have become widely-used representatives for short-clip classification. The second family is transformer-based video models: TimeSformer factorizes attention over space and time Bertasius et al. [2021], Video Swin extends shifted-window attention to video Liu et al. [2022], and VideoMAE pretrains video representations via masked autoencoding Tong et al. [2022]. The third uses vision-language models as frozen visual feature extractors, leveraging their broad semantic coverage; Qwen2.5-VL is a representative open VLM in this family Bai et al. [2025]. Our experiments compare a frozen VLM (Qwen2.5-VL) and a fine-tuned task-specific 3D-CNN (ResNet3D) on the same HAR benchmark, letting us measure how much of the vision-side accuracy comes from broad semantic priors as opposed to action-specific motion features learned on Kinetics.

2.3 Phase-aware and dynamic gating

Mixture-of-Experts architectures have produced strong results for multimodal time-series outside of HAR. MELON applies multimodal MoE with spectral-temporal fusion to long-term mobility estimation in critical care monitoring, providing evidence that expert routing can handle temporal multimodal signals at scale Zhang et al. [2024]. In visual-inertial odometry, MA-EVIO adaptively adjusts fusion weights based on motion dynamics, supporting the idea that motion-aware gating can outperform static fusion in high-dynamic regimes Shahraki et al. [2025]. Cross-modal alignment

between vision and IMU has been studied directly: C3T reports that strict cross-modal temporal synchronization is hard to achieve in practice Kamboj et al. [2024]. Cross-modal attention frameworks originally developed for vision-language tasks, notably ViLBERT Lu et al. [2019], have been adopted as a generic mechanism for fusing two streams. The broader literature on temporal action localization Wang et al. [2024] provides methodological background for identifying phase boundaries within an action, although these methods target localization as the end task rather than per-token gating for multimodal fusion.

Position of this work. Our framework builds on this literature with two specific choices. First, the fusion weight $\alpha(t)$ is predicted at the IMU token rate rather than at the sample or segment level, so that a single action can shift the trusted modality across its phases. Second, the input to the arbitrator is a small set of physically interpretable biomechanical quantities derived from the raw IMU signal (acceleration magnitude, a jerk-like second derivative, and a kinetic-energy rate), rather than learned latent features. To our knowledge, this combination of per-token granularity with a physics-grounded gating input has not been used for RGB-IMU fusion in HAR.

3 Problem Statement

Setup. We study multimodal human activity recognition (HAR) from synchronized RGB video and inertial measurement unit (IMU) data. An RGB video is $\mathbf{X}_v \in \mathbb{R}^{T_v \times H \times W \times 3}$ with T_v frames at 15 fps. The co-recorded IMU sequence is $\mathbf{X}_i \in \mathbb{R}^{T_i^{\text{raw}} \times 6}$, with T_i^{raw} timesteps and six channels (3-axis accelerometer and 3-axis gyroscope at 50 Hz). Each sample has a label $y \in \{1, \dots, C\}$. Given a training set $\{(\mathbf{X}_v^{(n)}, \mathbf{X}_i^{(n)}, y^{(n)})\}_{n=1}^N$, we want a classifier $f_\theta(\mathbf{X}_v, \mathbf{X}_i) \rightarrow \hat{y}$ that maximizes top-1 accuracy on a subject-disjoint test split.

Temporal heterogeneity. An action sequence is not equally informative across its duration. In a tennis serve or a throw, the preparation phase carries pose and scene cues that video reads well, while the high-velocity execution phase produces sharp accelerometer transients that the IMU captures more directly. Existing fusion methods do not contradict this picture, but they generally do not structure their fusion around it. Gating weights are usually predicted from learned latent features that combine both modalities, or are held constant within a sample. Our framework takes the biomechanical phase as the explicit input to fusion. Formally, let Acc_k^m denote the top-1 accuracy of modality $m \in \{\text{VISION}, \text{IMU}\}$ when a classifier is trained and tested only on the k -th normalized temporal segment of each sample:

$$\text{Acc}_k^m = \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{(\mathbf{x}, y) \in \mathcal{D}_{\text{test}}} \mathbf{1}[f_k^m(\mathbf{x}_k) = y], \quad k \in \{1, \dots, K\}. \quad (1)$$

We say the data exhibits temporal heterogeneity when $\text{Acc}_k^{\text{VISION}}$ and $\text{Acc}_k^{\text{IMU}}$ diverge non-trivially across k . Whether this holds in practice is an empirical question, taken up in Sec. ??.

Phase-gated fusion. Let $\mathbf{v}(t), \mathbf{i}(t) \in \mathbb{R}^d$ be temporally aligned vision and IMU token features at the t -th IMU token, with $t = 1, \dots, T_i$ and $T_i \ll T_i^{\text{raw}}$ the temporal resolution of the IMU encoder. Define a per-token fusion weight $\alpha \in [0, 1]^{T_i}$. The fused representation is

$$\mathbf{z}(t) = \alpha(t) \mathbf{v}(t) + (1 - \alpha(t)) \mathbf{i}(t), \quad t = 1, \dots, T_i, \quad (2)$$

and the classifier head g operates on its temporal mean, $\hat{y} = g\left(\frac{1}{T_i} \sum_t \mathbf{z}(t)\right)$. The standard early-fusion and late-fusion settings correspond to a constant α , so this formulation contains them as special cases.

Phase arbitrator. How α is produced is the key design choice of our framework. Rather than learning it from latent features through cross-modal attention, we predict it from a small set of biomechanical quantities derived from the raw IMU signal. A parameter-free feature map $\phi : \mathbb{R}^{T_i^{\text{raw}} \times 6} \rightarrow \mathbb{R}^{T_i^{\text{raw}} \times 3}$ extracts, at every timestep, the acceleration magnitude $|a(t)|$, its second time derivative $|\ddot{a}|(t)$ (a jerk-like quantity), and the kinetic-energy rate $\frac{d}{dt}|a(t)|^2$. The arbitrator is

$$\alpha = \sigma\left(\text{ArbMLP}(\text{Enc}(\text{Pool}_{T_i}(\phi(\mathbf{X}_i))))\right), \quad (3)$$

PG-MoE Architecture

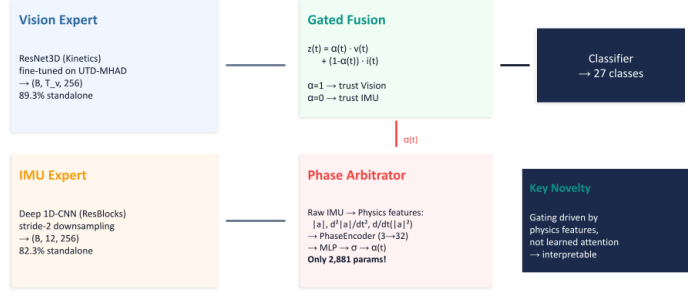


Figure 1: PG-MoE architecture. Vision Expert (ResNet3D) and IMU Expert (Deep 1D-CNN) produce temporal tokens. The Phase Arbitrator extracts physics features from raw IMU to produce per-token gating weights $\alpha(t)$.

where Pool_{T_i} matches the IMU encoder rate and Enc, ArbMLP are lightweight 1×1 convolutional networks. The full arbitrator has 2,881 trainable parameters.

4 Proposed Method

4.1 Overall Architecture

PG-MoE consists of four components (Figure 1): (1) a Vision Expert encoding RGB video into tokens $\mathbf{v} \in \mathbb{R}^{T_v \times d}$, (2) an IMU Expert encoding inertial data into tokens $\mathbf{i} \in \mathbb{R}^{T_i \times d}$ with $d = 256$, (3) a Phase Arbitrator producing per-token gating weights $\alpha(t) \in [0, 1]^{T_i}$ from raw IMU physics features, and (4) a gated fusion module. Vision tokens are aligned to T_i via adaptive average pooling, and the fused representation is $\mathbf{z}(t) = \alpha(t) \mathbf{v}(t) + (1 - \alpha(t)) \mathbf{i}(t)$, mean-pooled for classification. The gating is driven entirely by physics-derived features, making it interpretable.

4.2 Vision Expert

ResNet3D. We use r3d_18 Tran et al. [2018b] pretrained on Kinetics-400. Input volumes (3, 60, 112, 112) pass through five 3D convolutional stages, followed by spatial pooling, a projection layer (512 $\rightarrow$ 256), and a 2-layer temporal transformer. Two-stage training (freeze then fine-tune) achieves **89.3%**, up from the midterm’s 72.8%.

Qwen2.5-VL. We also evaluate frozen Qwen2.5-VL-3B Bai et al. [2025] tokens as an alternative backbone. Each frame produces 64 spatial tokens of 2048 dimensions; after spatial pooling and a temporal transformer, the best configuration achieves only **58.6%**. The 30.7-point gap versus ResNet3D reflects a pretraining mismatch: Qwen encodes *semantic* features (objects, layout) while ResNet3D encodes *motion* features (spatiotemporal dynamics from Kinetics-400).

The Vision Expert outputs $\mathbf{v} \in \mathbb{R}^{T_v \times 256}$ for downstream fusion.

4.3 IMU Expert

The IMU expert maps a raw 6-channel signal $\mathbf{X}_i \in \mathbb{R}^{6 \times T_i^{\text{raw}}}$ (with $T_i^{\text{raw}} = 192$ at 50 Hz) to a sequence of $T_i = 12$ tokens of dimension $d = 256$. We use a one-dimensional convolutional network with residual blocks and batch normalization. A 7-tap stride-2 stem (6 channels to 64) is followed by three residual blocks, each performing stride-2 temporal downsampling so that the temporal dimension shrinks $192 \rightarrow 96 \rightarrow 48 \rightarrow 24 \rightarrow 12$ and the channel count grows $64 \rightarrow 128 \rightarrow 256 \rightarrow 256$. Each residual block contains two convolutions with batch norm and ReLU activations, plus a 1×1 projection on the shortcut when the channel count or stride changes.

A central design choice is that the IMU expert does not collapse the temporal axis. Pooling to a single vector before fusion would force every downstream module, including the gated fusion and the phase arbitrator, to operate at sample granularity, which is incompatible with the per-token formulation of Eq. 2. The encoder instead emits the full 12-token sequence and lets the fusion module decide how to aggregate.

Trained as a standalone classifier with temporal mean-pooling and a single linear head, this encoder reaches 82.3% top-1 accuracy on the UTD-MHAD test split (Sec. ??). The same encoder is reused as a frozen feature extractor inside the full PG-MoE.

4.4 Phase Arbitrator

The phase arbitrator produces the per-token fusion weight $\alpha \in [0, 1]^{T_i}$ from the raw IMU signal. It is the defining component of PG-MoE and is implemented as a three-stage module: a deterministic biomechanical feature map followed by a small encoder and an MLP head.

Stage 1: Biomechanical features. A parameter-free feature map $\phi : \mathbb{R}^{6 \times T_i^{\text{raw}}} \rightarrow \mathbb{R}^{3 \times T_i^{\text{raw}}}$ extracts three quantities from the 3-axis accelerometer channels $\mathbf{a}(t) = (a_x(t), a_y(t), a_z(t))$ at every timestep:

$$|a(t)| = \sqrt{a_x(t)^2 + a_y(t)^2 + a_z(t)^2}, \quad (4)$$

$$|\ddot{a}|(t) = |a|(t) - 2|a|(t-1) + |a|(t-2), \quad (5)$$

$$\dot{E}(t) = |a(t)|^2 - |a(t-1)|^2. \quad (6)$$

The first quantity, the acceleration magnitude, captures the instantaneous effort. The second, a discrete second-difference of the magnitude, is a jerk-like signal that responds sharply at transitions between biomechanical phases such as the start of a downswing. The third, the rate of change of kinetic-energy-like quantity, amplifies periods of high power output. These three quantities are concatenated channel-wise into $\phi(\mathbf{X}_i)$ and require no learning.

Stage 2: Phase encoder. $\phi(\mathbf{X}_i)$ is first downsampled to T_i tokens by adaptive average pooling along the temporal axis, matching the IMU expert resolution. The pooled sequence is then passed through two 1×1 convolutions $3 \rightarrow 64 \rightarrow 32$ with ReLU activations. Because the convolutions are point-wise in time, each output channel at token t depends only on the three biomechanical features at the same token. This keeps the gating signal localized to a single phase rather than mixed across the action.

Stage 3: Arbitrator MLP. Two further 1×1 convolutions $32 \rightarrow 16 \rightarrow 1$ followed by a sigmoid produce $\alpha \in [0, 1]^{T_i}$. The full arbitrator (Stages 2 and 3 combined) contains 2,881 trainable parameters.

Design rationale. Two choices set the arbitrator apart from previous dynamic-fusion modules. First, the input is a small set of physically interpretable biomechanical quantities derived from the raw IMU, rather than a latent feature vector. Second, the output is computed per IMU token, so a single action can shift the trusted modality across its preparation and execution phases. The combination produces a gating signal that can be inspected directly in physical units, and it does so with two orders of magnitude fewer parameters than a typical attention-based fusion block.

4.5 Gated Fusion and Training Objective

Fusion. The vision expert produces a sequence $\mathbf{V} \in \mathbb{R}^{T'_v \times d}$ at its own temporal rate T'_v , while the IMU expert produces $\mathbf{I} \in \mathbb{R}^{T_i \times d}$. To apply the per-token fusion of Eq. 2, we align the vision sequence to the IMU rate by one-dimensional adaptive average pooling along the temporal axis, yielding $\mathbf{V}' \in \mathbb{R}^{T_i \times d}$. The fused representation at token t is

$$\mathbf{z}(t) = \alpha(t) \mathbf{v}'(t) + (1 - \alpha(t)) \mathbf{i}(t), \quad t = 1, \dots, T_i. \quad (7)$$

The classifier head is a LayerNorm followed by a single linear layer applied to the temporal mean $\bar{\mathbf{z}} = \frac{1}{T_i} \sum_t \mathbf{z}(t)$, producing the C -way logits.

Loss. UTD-MHAD has only 431 training samples spread across 27 classes and the per-class hardness varies substantially. We use focal loss Lin et al. [2017] with focusing parameter $\gamma = 2$:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N} \sum_{n=1}^N (1 - p_{y_n})^\gamma \log p_{y_n}, \quad (8)$$

where p_{y_n} is the softmax probability assigned to the ground-truth class. We add two auxiliary terms on the arbitrator output. A variance term $\mathcal{L}_{\text{var}} = -\frac{1}{N} \sum_n \text{Var}_t[\alpha^{(n)}(t)]$ encourages within-sample temporal variation, preventing α from collapsing to a constant value within an action. A balance term $\mathcal{L}_{\text{bal}} = (\bar{\alpha} - 0.5)^2$, computed on the batch-mean of α , prevents α from collapsing to either extreme of $\{0, 1\}$, which would reduce the model to a single-modality classifier. The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_v \mathcal{L}_{\text{var}} + \lambda_b \mathcal{L}_{\text{bal}}, \quad (9)$$

with $\lambda_v = 0.05$ and $\lambda_b = 0.1$.

Optimization. We train the joint model with AdamW Loshchilov and Hutter [2019] under a cosine learning-rate schedule. The pretrained IMU and vision encoders use a small learning rate of 5×10^{-5} to preserve their standalone representations, while the newly initialized vision projection, gated fusion head, and phase arbitrator use a larger learning rate of 10^{-3} . Gradient norm is clipped at 1.0. At inference time the arbitrator is evaluated once per sample, producing the full α sequence in a single forward pass with no post-processing. Specific batch sizes, epoch counts, and hardware configuration are given in Sec. 5.

5 Experimental Methodology

Dataset. We evaluate on UTD-MHAD Chen et al. [2015]: 27 action classes, 8 subjects, 861 sequences. We use **RGB + IMU only**—a modality pair available on any smartphone—rather than skeleton and depth, which require a Kinect sensor and achieve 98%+ on this benchmark. The RGB+IMU setting is more practical but correspondingly more challenging.

Evaluation. Standard subject-disjoint split: subjects $\{1, 3, 5, 7\}$ train (431 samples), $\{2, 4, 6, 8\}$ test (430 samples). Primary metric: top-1 accuracy. We also report an *oracle* upper bound—accuracy if each sample is routed to whichever expert is correct.

Baselines. We compare nine methods using the same expert backbones: (1–3) three single-modality baselines (IMU 1D-CNN, ResNet3D, Qwen2.5-VL); (4) Early Fusion (concatenate mean-pooled features); (5) Late Fusion (average logits, no learnable parameters); (6) Cross-Attention without gating (2-layer Co-TRM); (7–9) three PG-MoE variants (token-level, logit-level, and fine-tuned). All fusion methods share the same frozen or partially-unfrozen experts to isolate the fusion mechanism.

Implementation. All experiments use Google Colab (T4/A100) with PyTorch. RGB: 112×112 , 60 frames. IMU: 192 timesteps at 50 Hz. Vision Expert trains in two stages: 80 epochs frozen backbone ($\text{lr} = 3 \times 10^{-4}$), then 40 epochs full fine-tune ($\text{lr} = 5 \times 10^{-5}$). IMU Expert: 80 epochs, $\text{lr} = 10^{-3}$. PG-MoE: 60–80 epochs with differentiated learning rates (experts 3×10^{-5} , heads 5×10^{-4}), focal loss ($\gamma = 2$). Qwen2.5-VL tokens extracted on A100, cached to disk (13.54 GB). Full hyperparameter details in Appendix.

6 Results and Discussion

6.1 Single-Modality Results

Table 1 presents single-modality results. Upgrading the midterm IMU baseline from a shallow 2-layer CNN to a deep residual 1D-CNN with stride-2 downsampling improves accuracy from 67.9% to 82.3%. The Vision Expert (ResNet3D) reaches 89.3% with a temporal transformer and two-stage fine-tuning, up from the midterm’s 72.8%.

The frozen Qwen2.5-VL achieves only 58.6% despite 668M parameters. We tested three pooling strategies (global mean: 28.1%, temporal attention: 53.3%, temporal transformer: 58.6%); the 30.7-point gap versus ResNet3D confirms that VLM encoders capture semantics rather than the motion dynamics critical for HAR.

Table 1: Single-modality results on UTD-MHAD.

Modality	Model	Parameters	Accuracy
IMU	1D-CNN (midterm)	46.6K	67.9%
IMU	Deep 1D-CNN (final)	1.1M	82.3%
Vision	Qwen2.5-VL (frozen)	668M	58.6%
Vision	ResNet3D (fine-tuned)	34.9M	89.3%

Table 2: Comparison of fusion methods on UTD-MHAD (27 classes).

Method	Type	Params	Accuracy
IMU-only	Single	1.1M	82.3%
Vision-only	Single	34.9M	89.3%
Early Fusion	Fusion	14.9K	90.2%
Late Fusion	Fusion	0	92.3%
Cross-Attn (no gating)	Fusion	2.1M	89.8%
PG-MoE (token-level)	Ours	10.3K	90.2%
PG-MoE (fine-tuned)	Ours	2.2M	91.4%
Oracle upper bound	—	—	98.4%

6.2 Fusion Method Comparison

Table 2 compares fusion methods. All multimodal approaches exceed the single-modality best (89.3%), confirming RGB–IMU complementarity. Late Fusion (parameter-free logit averaging) achieves 92.3%, outperforming all learned methods. We attribute this to the small training set: with 431 samples, learnable fusion modules risk overfitting, while averaging acts as an implicit ensemble. Cross-modal attention (2.1M parameters) achieves only 89.8%, suggesting that on small datasets the additional capacity dilutes pretrained representations. The best PG-MoE variant (91.4%, fine-tuned) partially unfreezes expert layers with differentiated learning rates; it does not surpass Late Fusion in accuracy but provides interpretable $\alpha(t)$ that Late Fusion lacks.

6.3 Per-Class Complementarity Analysis

Table: Per-class complementarity (selected classes)

Table 3 shows per-class accuracy for both experts. The two exhibit strikingly different error profiles: vision achieves 100% on 12 classes (explosive actions like boxing, baseball swing) but drops on visually subtle ones (throw: 56%); IMU achieves 100% on 10 classes (rhythmic actions like wave, draw circle) but struggles with brief transients (tennis serve: 50%). The largest gaps are tennis serve (+44% vision) and wave (+25% IMU).

The oracle (correct if either expert is right) reaches **98.4%**: only 7/430 samples fool both experts. The 6.1-point gap to Late Fusion (92.3%) represents 26 recoverable errors. The per-class structure shows that optimal routing is class-conditional—requiring more training data than UTD-MHAD provides, which explains why parameter-free averaging remains competitive.

6.4 Phase Arbitrator Analysis

Before training, the randomly initialized arbitrator outputs $\alpha(t) = 0.527 \pm 0.001$ across all actions and timesteps. After joint training α shifts modestly: the per-class mean varies, but the within-sample temporal variance of $\alpha(t)$ remains small, and the curves for high-dynamic actions like throwing and tennis serve do not show the sharp preparation-to-execution transition that the temporal-heterogeneity hypothesis would predict.

Three factors plausibly contribute. UTD-MHAD provides only 431 training samples across 27 classes, giving each class-specific gating profile very few examples to fit. Actions are short (about 2–4 s, only $T_i = 12$ tokens after encoding), leaving little room for phase transitions to emerge within a sample.

Table 3: Per-class accuracy comparison (selected classes showing strongest complementarity). Full 27-class table in Appendix.

Action	IMU	Vision	Δ
Tennis serve	50%	94%	+44%
Throw	25%	56%	+31%
Boxing	69%	100%	+31%
Pickup & throw	69%	100%	+31%
Baseball swing	81%	100%	+19%
Wave	100%	75%	−25%
Draw circle (CW)	100%	75%	−25%
Draw circle (CCW)	100%	75%	−25%
Arm curl	88%	69%	−19%
Clap	100%	100%	0%
Walk	100%	100%	0%
Squat	100%	100%	0%
Overall	82.3%	89.3%	
Oracle		98.4%	

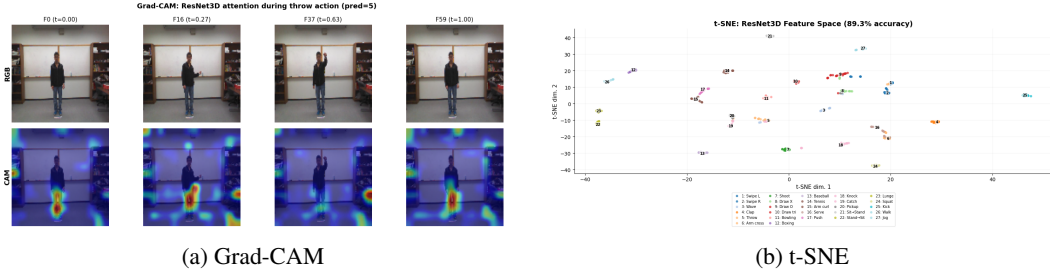


Figure 2: (a) Grad-CAM shows ResNet3D attends to person but also background edges. (b) t-SNE shows well-separated clusters for 27 action classes.

And the IMU expert already reaches 82.3% standalone, so once its logits dominate the fusion loss, the gradient signal pushing α toward a phase-dependent shape is weak.

Two observations support reading this as a data-scale limit rather than an architectural one. Late Fusion (92.3%) edges out PG-MoE (91.4%), exactly as one would expect if the arbitrator cannot extract phase-dependent advantage at this size. And the 98.4% oracle ceiling confirms that strong per-class complementarity exists in the data, even when the trained arbitrator cannot recover it. We therefore read PG-MoE as a framework whose value should grow with dataset scale.

6.5 Visualization Analysis

Grad-CAM. Grad-CAM at layer3 (14×14) of ResNet3D (Figure 2a) shows the model attends to the person’s body but also activates on background structures (bookshelf edges, wall boundaries) consistent across all UTD-MHAD recordings. This partial scene bias motivates IMU as a scene-independent complement.

t-SNE. t-SNE of mean-pooled ResNet3D features (Figure 2b) shows well-separated clusters for 27 classes, consistent with 89.3% accuracy. Overlaps between visually similar pairs (swipe left/right, sit-to-stand/ stand-to-sit) suggest these differ primarily in temporal ordering—a distinction better captured by IMU.

Phase features. Figure 3 visualizes the three biomechanical signals that drive the Phase Arbitrator on a throw action. The acceleration magnitude $|a(t)|$ peaks during release, the jerk-like second derivative $|\ddot{a}(t)|$ spikes sharply at the boundaries between preparation, release, and follow-through,

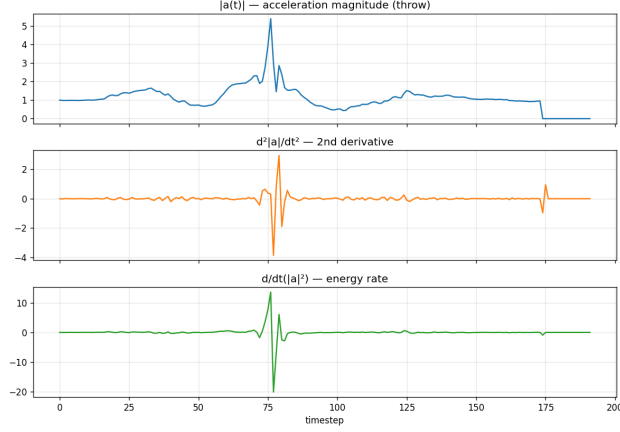


Figure 3: Biomechanical features driving the Phase Arbitrator on a throw action: acceleration magnitude $|a(t)|$ (top), jerk-like second derivative $\ddot{|a|}(t)$ (middle), and energy rate $\dot{E}(t)$ (bottom). Sharp transitions between preparation, release, and follow-through are visible.

and the energy-rate signal $\dot{E}(t)$ amplifies the brief window of peak power output. These curves give the arbitrator a physically meaningful, class-distinctive input. As discussed in Sec. 6.4, the trained arbitrator does not fully exploit this structure at UTD-MHAD scale, but the input itself is rich and well-localized in time.

7 Conclusion and Future Directions

We presented PG-MoE, a phase-gated mixture-of-experts framework for multimodal HAR that gates RGB–IMU fusion using biomechanical quantities derived from raw IMU signals. The Phase Arbitrator requires only 2,881 parameters and produces interpretable per-token weights grounded in acceleration magnitude, jerk, and kinetic energy rate.

Evaluated on UTD-MHAD with RGB + IMU (a practical but challenging modality pair), we compared nine fusion strategies. All fusion methods surpass the best single-modality result (89.3%), confirming modality complementarity. Late Fusion achieves 92.3% and PG-MoE 91.4%, while an oracle analysis reveals a 98.4% upper bound—26 recoverable errors that neither method currently exploits. We additionally show that frozen VLM features (Qwen2.5-VL, 58.6%) substantially underperform task-specific video models, highlighting the importance of motion-aware pretraining for HAR.

The main limitation is dataset scale: 431 training samples proved insufficient for learned gating to outperform parameter-free averaging, and the Arbitrator’s $\alpha(t)$ converged near 0.5. This is itself an informative finding: the per-class analysis shows that optimal routing is class-conditional, requiring more data than UTD-MHAD provides. Future work should evaluate PG-MoE on larger datasets (ActionSense, NTU RGB+D), explore tri-modal fusion with skeleton, and test egocentric deployment with smartwatch IMU and phone camera.

References

- Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Haiyang Xu, Jinze Bai, Junyang Lin, Chang Zhou, and Jingren Zhou. Qwen2.5-VL technical report. *arXiv preprint arXiv:2502.13923*, 2025.
- Gedas Bertasius, Heng Wang, and Lorenzo Torresani. Is space-time attention all you need for video understanding? In *International Conference on Machine Learning (ICML)*, 2021.
- Joao Carreira and Andrew Zisserman. Quo vadis, action recognition? a new model and the Kinetics dataset. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6299–6308, 2017.

- Chen Chen, Roozbeh Jafari, and Nasser Kehtarnavaz. UTD-MHAD: A multimodal dataset for human action recognition utilizing a depth camera and a wearable inertial sensor. In *IEEE International Conference on Image Processing (ICIP)*, pages 168–172, 2015. doi: 10.1109/ICIP.2015.7350781.
- Christoph Feichtenhofer, Haoqi Fan, Jitendra Malik, and Kaiming He. SlowFast networks for video recognition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.
- Ziqi Gao, Yuntao Wang, Jianguo Chen, Junliang Xing, Shwetak Patel, Xin Liu, and Yuanchun Shi. Mmtsa: Multi-modal temporal segment attention network for efficient human activity recognition. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 7(3):1–26, 2023. doi: 10.1145/3610872.
- Abhi Kamboj, Anh Duy Nguyen, and Minh N. Do. C3T: Cross-modal transfer through time for sensor-based human activity recognition. *arXiv preprint*, 2024. Please verify venue / arXiv ID.
- Will Kay, Joao Carreira, Karen Simonyan, Brian Zhang, Chloe Hillier, Sudheendra Vijayanarasimhan, Fabio Viola, Tim Green, Trevor Back, Paul Natsev, Mustafa Suleiman, and Andrew Zisserman. The Kinetics human action video dataset. *arXiv preprint arXiv:1705.06950*, 2017.
- Ismail Lamaakal, Chaymae Yahyati, Yassine Maleh, Khalid El Makkaoui, Ibrahim Ouahbi, Ahmed A. Abd El-Latif, Mariam Zomorodi, and Basma Abd El-Rahiem. A tiny inertial transformer for human activity recognition via multimodal knowledge distillation and explainable AI. *Scientific Reports*, 15(1):42335, 2025.
- Timin Li, Dongmei Li, Yuepeng Chen, Yihe Wang, Xuefeng Feng, Ye Ma, Dongwei Liu, Ji Wu, and Chenyi Guo. DG-IVF: A dual-gated IMU-vision fusion method for efficient HAR. In *Neural Information Processing (ICONIP)*, volume 2754 of *Communications in Computer and Information Science*, pages 406–420. Springer, 2023.
- Paul Pu Liang, Yiwei Lyu, Xiang Fan, Zetian Wu, Yun Cheng, Jason Wu, Leslie Chen, Peter Wu, Michelle A. Lee, Yuke Zhu, Ruslan Salakhutdinov, and Louis-Philippe Morency. Multibench: Multiscale benchmarks for multimodal representation learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- Paul Pu Liang, Amir Zadeh, and Louis-Philippe Morency. Foundations and trends in multimodal machine learning: Principles, challenges, and open questions. *arXiv preprint arXiv:2209.03430*, 2022.
- Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017.
- Ze Liu, Jia Ning, Yue Cao, Yixuan Wei, Zheng Zhang, Stephen Lin, and Han Hu. Video Swin transformer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.
- Jiasen Lu, Dhruv Batra, Devi Parikh, and Stefan Lee. ViLBERT: Pretraining task-agnostic vision-linguistic representations for vision-and-language tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- Mohsen Shahraki, Ahmed Elamin, and Ahmed El-Rabbany. MA-EVIO: A motion-aware approach to event-based visual-inertial odometry. *Sensors*, 25(23):7381, 2025.
- Zhan Tong, Yibing Song, Jue Wang, and Limin Wang. VideoMAE: Masked autoencoders are data-efficient learners for self-supervised video pre-training. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Du Tran, Heng Wang, Lorenzo Torresani, Jamie Ray, Yann LeCun, and Manohar Paluri. A closer look at spatiotemporal convolutions for action recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6450–6459, 2018a.

- Du Tran, Heng Wang, Lorenzo Torresani, Jamie Ray, Yann LeCun, and Manohar Paluri. A closer look at spatiotemporal convolutions for action recognition. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6450–6459, 2018b. doi: 10.1109/CVPR.2018.00675.
- Binglu Wang, Yongqiang Zhao, Le Yang, Teng Long, and Xuelong Li. Temporal action localization in the deep learning era: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(4):2171–2190, 2024.
- Sanju Xaviar, Xin Yang, and Omid Ardakanian. Centaur: Robust multimodal fusion for human activity recognition. *Sensors*, 24(11):18578–18591, 2024.
- Zihui Xue and Radu Marculescu. Dynamic multimodal fusion. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.
- Jiaqing Zhang, Miguel Contreras, Jessica Sena, Andrea Davidson, Yuanfang Ren, Ziyuan Guan, Tezcan Ozrazgat-Baslanti, Tyler J. Loftus, Subhash Nerella, Azra Bihorac, and Parisa Rashidi. MELON: Multimodal mixture-of-experts with spectral-temporal fusion for long-term mobility estimation in critical care. In *Machine Learning for Healthcare (MLHC)*, 2024.